United States Patent Application Publication	20250281957
Kind Code	A1
Publication Date	September 11, 2025
Inventor(s)	Nakamura; Hidenori et al.

COOLING DEVICE AND HEAT TREATMENT APPARATUS INCLUDING THE SAME

Abstract

A cooling device includes a loading section, an unloading section, plural cooling rollers, and a pair of support plates. The pair of support plates supports the plural cooling rollers. The support plates in a pair are opposed to each other to sandwich a space in which the cooling rollers are arranged in a staggered manner. The pair of support plates includes bearings that support respective roller shafts of the plural cooling rollers. A conveyance route for the workpiece is set so as to bring a first surface of a workpiece into contact with a cooling roller displaced toward a first side and bring a second side of the workpiece into contact with a cooling roller displaced toward a second side, among the plural cooling rollers.

Inventors: **Nakamura; Hidenori (Nagoya-shi, JP), Ohi; Hideaki (Nagoya-shi, JP), Iida; Masaki (Nagoya-shi, JP)**

Applicant: **Noritake Co., Limited** (Nagoya-shi, JP)

Family ID: **1000008547966**

Assignee: **Noritake Co., Limited (Nagoya-shi, JP)**

Appl. No.: **19/071061**

Filed: **March 05, 2025**

Foreign Application Priority Data

JP	2024-033594	Mar. 06, 2024

Publication Classification

Int. Cl.: **B21B45/02** (20060101); **B21B1/26** (20060101)

U.S. Cl.:

CPC **B21B45/0209** (20130101); **B21B1/26** (20130101);

Background/Summary

CROSS REFERENCE TO RELATED APPLICATION

[0001] This application claims the benefit of priority to Japanese Patent Application No. 2024-033594 filed on Mar. 6, 2024, the entire contents of which are hereby incorporated herein by reference.

BACKGROUND OF THE DISCLOSURE

[0002] The present disclosure relates to a cooling device and a heat treatment apparatus including the same.

[0003] Japanese Patent Publication No. 7285360 discloses a heat treatment apparatus that includes an unwinding unit, a heat treatment unit, a cooling unit, and a winding unit. The cooling unit is provided with plural cooling rollers. In the cooling unit, a strip-shaped workpiece is cooled and conveyed while being hung on the cooling rollers.

SUMMARY

[0004] The present inventors intend to improve the workability of hanging a strip-shaped workpiece on plural cooling rollers.

[0005] A cooling device disclosed herein includes a loading section, an unloading section, a plurality of cooling rollers, and a pair of support plates. The loading section loads a strip-shaped workpiece. The unloading section unloads the workpiece. The pair of support plates supports the plurality of cooling rollers. The cooling rollers are arranged in a staggered manner while being aligned with each other in an axial direction, with positions thereof sequentially displaced along a height direction. The support plates in a pair are opposed to each other to sandwich a space in which the cooling rollers are arranged in the staggered manner. The pair of support plates includes bearings that support respective roller shafts of the plural cooling rollers. A conveyance route for the workpiece is set such that the workpiece is loaded from the loading section, sequentially hung around the plurality of cooling rollers along the height direction, and then unloaded from the unloading section. The conveyance route for the workpiece is set so as to bring a first surface of the workpiece into contact with a cooling roller displaced toward a first side and bring a second surface of the workpiece into contact with a cooling roller displaced toward a second side, among the plural cooling rollers. Such a cooling device improves the workability of hanging the strip-shaped workpiece on the plurality of cooling rollers.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0006] FIG. **1** is a schematic diagram illustrating a heat treatment apparatus **10** and a cooling device **50**.

[0007] FIG. **2** is a schematic diagram illustrating a support structure of cooling rollers **52**.

[0008] FIG. **3** is a schematic diagram illustrating the support structure of the cooling rollers **52**.

[0009] FIG. **4** is a schematic diagram illustrating a first support portion **56***a*.

DETAILED DESCRIPTION OF THE EMBODIMENTS

[0010] One embodiment in the present disclosure will be described in detail below with reference to the drawings. In the following drawings, members and portions that have the same actions are denoted by the same symbols. The dimensional relationships (length, width, thickness, etc.) in each drawing do not reflect the actual dimensional relationships. The directions of up, down, left, right, front, and back are represented are represented by arrows U, D, L, R, F, and Rr, respectively, in the figures. Here, the directions of up, down, left, right, front, and back are only provided for convenience of explanation and do not limit the present disclosure unless otherwise specifically mentioned.

Heat Treatment Apparatus **10**

[0011] FIG. **1** is a schematic diagram illustrating a heat treatment apparatus **10** and a cooling device **50**. The heat treatment apparatus **10** is equipment for performing heat treatment on a strip-shaped (sheet-shaped) workpiece A. In this embodiment, the heat treatment apparatus **10** is a device for continuously drying the strip-shaped workpiece while conveying it in a so-called roll-to-roll system. The workpiece A is not particularly limited as long as it is a strip-shaped object, such as an electrode sheet for secondary batteries that has an electrode material applied to both sides of a sheet base, a Flexible Copper Clad Laminate (FCCL), or a polyimide sheet. The heat treatment apparatus **10** can be used to process a variety of strip-shaped (sheet-shaped) workpieces.

[0012] As illustrated in FIG. **1**, the heat treatment apparatus **10** includes an unwinding unit **30**, a heat treatment unit **40**, a cooling unit **50**, and a winding unit **60**. The heat treatment apparatus **10** includes conveyance devices **20** and **22**. The cooling device **50** disclosed herein is used as the cooling unit **50** in the heat treatment apparatus **10**. The strip-shaped workpiece A is conveyed and processed through the unwinding unit **30**, the heat treatment unit **40**, the cooling unit **50**, and the winding unit **60** in that order. The workpiece A is unwound from an unwinding roll A**1** provided in the unwinding unit **30**, heat-treated in the heat treatment unit **40**, cooled in the cooling unit **50**, and then wound onto a winding roll A**2** provided in the winding unit **60**.

Conveyance Devices **20** and **22**

[0013] The conveyance devices **20** and **22** are devices that convey the workpiece A. The workpiece A is conveyed along a predetermined conveyance route. The conveyance devices **20** and **22** are devices that rotationally drive an unwinding shaft **32**, on which the unwinding roll A**1** of the unwinding unit **30** is attached, and a winding shaft **62**, on which the winding roll A**2** of the winding unit **60** is attached, respectively. In this embodiment, motors are used as the conveyance devices **20** and **22**. Each of the conveyance devices **20** and **22** can be constituted of a device that controls the conveyance of the workpiece A. For example, a motor and an inverter may be used as the conveyance devices **20** and **22**, and particularly a servo motor or the like may be used. The conveyance devices **20** and **22** may include a device that controls the tension applied to the workpiece A. For example, a powder clutch may be used as the device for controlling the tension. The conveyance devices **20** and **22** can be implemented using a set of devices where one controls the conveyance speed and another controls the tension, operating in cooperation with each other.

[0014] The unwinding shaft **32** is connected to the conveyance device **20**. The unwinding shaft **32** is rotationally driven by the conveyance device **20**, thereby unwinding the workpiece A from the unwinding roll A**1**. The winding shaft **62** is connected to the conveyance device **22**. The winding shaft **62** is rotationally driven by the conveyance device **22**, thereby winding the workpiece A onto the winding roll A**2**. The conveyance devices **20** and **22** may be installed in respective atmospheric boxes provided in spaces enclosed by outer walls **31** and **61**, respectively. The conveyance devices **20** and **22** may be provided outside the outer walls **31** and **61**, respectively. The heat treatment apparatus **10** can be configured to convey the workpiece A at a high speed in order to improve the processing efficiency of the workpiece A. Although not particularly limited, the conveyance speed of the workpiece A can be set to about 1 m/min to 200 m/min. In this embodiment, the conveyance speed of the workpiece A is set to about 100 m/min. In the heat treatment apparatus **10**, the conveyance speed of the workpiece A is controlled by a controller (not illustrated).

[0015] The controller controls the conveyance speed of the workpiece A, the tension applied to the workpiece A, and the like such that the workpiece A is conveyed according to a predetermined conveyance condition. The controller respectively controls an unwinding tension applied when the workpiece A is unwound, an in-furnace tension applied to the workpiece A being processed, and a winding tension applied when the processed workpiece A is wound up. The controller is connected to the conveyance devices **20** and **22**. The controller may be connected to a tension detection roller **35***b*, a feed roller **35***c*, a dancer roller **35***d*, a tension detection roller **65***c*, and the like. The controller feeds back the unwinding tension detected by the tension detection roller **35***b* to the conveyance device **20** and thereby controls the torque of the unwinding shaft **32**. Thus, the unwinding tension is adjusted. The controller also feeds back, to the dancer roller **35***d*, the in-furnace tension detected by the tension detection roller **35***b*, over which the workpiece A being processed is hung. The dancer roller **35***d* moves according to the detected in-furnace tension. Thus, the in-furnace tension is adjusted. The rotation speed of the feed roller **35***c* is controlled such that the position of the dancer roller **35***d* returns to a reference position with the in-furnace tension being constant. The controller feeds back the winding tension detected by the tension detection roller **65***c* to the conveyance device **22** and thereby controls the torque of the winding shaft **62**. Thus, the winding tension is adjusted.

Unwinding Unit **30**

[0016] The unwinding unit **30** is equipment that unwinds the workpiece A. The unwinding unit **30** houses the unwinding roll A**1** in a state where the workpiece A before the heat treatment is wound thereon. The unwinding unit **30** has the outer wall **31** that encloses internal equipment and the unwinding roll A**1**. The unwinding unit **30** includes therein the unwinding shaft **32** and plural rollers **35**. The unwinding shaft **32** is a shaft to which the unwinding roll A**1**, holding the workpiece A wound on it before heat treatment, is attached. In this embodiment, the unwinding shaft **32** is rotationally driven to unwind the workpiece A from the unwinding roll A**1** attached to the unwinding shaft **32**.

[0017] Within the space enclosed by the outer wall **31** of the unwinding unit **30**, the plural rollers **35** that set the conveyance route for the workpiece A are provided. The workpiece A unwound from the unwinding roll A**1** is hung around and sequentially passed through the plural rollers **35** in a predetermined order and then conveyed toward the heat treatment unit **40**. The plural rollers **35** include guide rollers **35***a*, a tension detection roller **35***b*, a feed roller **35***c*, and a dancer roller **35***d*. The tension detection roller **35***b* is a roller for detecting the tension applied to the workpiece A. A tension detector (not illustrated) is attached to the tension detection roller **35***b*. The dancer roller **35***d* is configured to be movable within a predetermined range. By moving the dancer roller **35***d*, the tension applied to the workpiece A is adjusted. The feed roller **35***c* is rotationally driven by a drive device (not illustrated). By controlling the rotation of the feed roller **35***c*, the position of the dancer roller **35***d* is adjusted.

Heat Treatment Unit **40**

[0018] The heat treatment unit **40** is equipment where the strip-shaped workpiece A is heat-treated while being conveyed. The heat treatment unit **40** is connected to the unwinding unit **30** via a connecting part **70**. The connecting part **70** is provided with an outlet for the unwinding unit **30** and an inlet for the heat treatment unit **40**. A path through which the workpiece A passes is formed in the connecting part **70**. The workpiece A is conveyed from the unwinding unit **30** to the heat treatment unit **40** through the connecting part **70**. The path for the workpiece A formed in the connecting part **70** is set to have dimensions slightly larger than the width and thickness of the workpiece A. This reduces the likelihood of interference between the atmosphere of the heat treatment unit **40** and that of the unwinding unit **30**.

[0019] Although a detailed illustration thereof is omitted, the heat treatment unit **40** may include an outer wall **41**, a heater, and guide rollers. The heat treatment unit **40** has therein a processing space where the unwound workpiece A is heat-treated while being conveyed. The outer wall **41** encloses the processing space in which the heater and the guide rollers are disposed.

[0020] The conveyance route along which the workpiece A is conveyed is set by the guide rollers. The guide roller is configured to rotate in a driven manner as the workpiece A is conveyed.

[0021] The heater is equipment for heating the workpiece A. The workpiece A is heat-treated by the heater while being conveyed in the conveyance route set by the guide rollers. Note that the configuration of the heat treatment unit **40**, including the guide rollers, the heater, and the like, is not particularly limited. The heat-treated workpiece A is conveyed out toward the cooling unit **50**.

Cooling unit **50**

[0022] The cooling unit **50** is equipment in which the workpiece A heat-treated in the heat treatment unit **40** is cooled while being conveyed. The cooling unit **50** is connected to the heat treatment unit **40** via a connecting part **72**. The cooling unit **50** includes a loading section **50***b*, an unloading section **50***c*, plural cooling rollers **52**, and a pair of support plates **53** and **54** (see FIGS. **2** and **3**). The cooling unit **50** also includes plural guide rollers **55** and an outer wall **51**.

[0023] The strip-shaped workpiece A is loaded from the loading section **50***b*. The loading section **50***b* is provided in the connecting part **72**. The workpiece A is unloaded from the unloading section **50***c*. The unloading section **50***c* is provided in a connecting part **74**. The loading section **50***b* and the unloading section **50***c* are positioned to be opposed to each other. The positional relationship between the loading section **50***b* and the unloading section **50***c* is not particularly limited. In the connecting part **72**, like the connecting part **70**, a passageway is formed through which the workpiece A passes. The configuration of the connecting part **72** can be the same as that of the connecting part **70**, and thus a detailed description thereof is omitted.

Cooling Roller **52**

[0024] The cooling roller **52** is a roller configured to allow a refrigerant to circulate therein. The workpiece A is cooled by contacting a surface of the cooling roller **52**. The cooling rollers **52** include a first cooling roller **52***a* and a second cooling roller **52***b*. The first cooling roller **52***a* is the cooling roller **52** around which the workpiece A is hung so as to bring its first surface A**3** into contact with the roller. The second cooling roller **52***b* is the cooling roller **52** around which the workpiece A is hung so as to bring its second surface A**4** into contact with the roller. In this embodiment, each cooling roller **52** is connected to a drive device (not illustrated). The cooling roller **52** rotates along the conveyance direction at a set conveyance speed. In this embodiment, the workpiece A is cooled to about room temperature in the cooling unit **50**. The temperature of the workpiece A to be cooled is not particularly limited.

[0025] Plural first cooling rollers **52***a* (four in this embodiment) are provided. In addition, plural second cooling rollers **52***b* (four in this embodiment) are also provided. The number of cooling rollers **52** is not particularly limited. However, for example, one first cooling roller **52***a* and one second cooling roller **52***b* may be provided.

[0026] The plural first cooling rollers **52***a* are arranged in a row at a predetermined pitch along the height direction at the front (outlet side) of the cooling unit **50**. The plural second cooling rollers **52***b* are arranged in a row at a predetermined pitch along the height direction at the rear (loading section **50***b* side) of the cooling unit **50**. A spacing between adjacent first cooling rollers **52***a* and a spacing between adjacent second cooling rollers **52***b* are each set to be narrower than an outer diameter of the cooling roller **52**. The arrangement of the first cooling rollers **52***a* and the second cooling rollers **52***b* is not particularly limited.

[0027] Here, the first cooling rollers **52***a* and the second cooling rollers **52***b* are arranged at the same pitch. The second cooling roller **52***b* is sequentially positioned by half a pitch higher than the first cooling roller **52***a*, as viewed from the bottom. Thus, the plural first cooling rollers **52***a* and the plural second cooling rollers **52***b* are arranged such that their heights are differentiated sequentially at the front and rear of the cooling unit **50**. In other words, the plural first cooling rollers **52***a* and the plural second cooling rollers **52***b* are arranged in a staggered manner. This allows the cooling unit **50** to have a structure elongated in the height direction. On the other hand, an exclusive area of the cooling unit **50** can be reduced to save space in a facility.

[0028] The outer wall **51** encloses a processing space **50***a* in which the plural cooling rollers **52** and the plural guide rollers **55** are disposed. In this embodiment, the plural cooling rollers **52** are provided in the cooling unit **50**. The plural cooling rollers **52** and the plural guide rollers **55** set the conveyance route along which the workpiece A is conveyed within the cooling unit **50**.

Guide Rollers **55**

[0029] The guide rollers **55** (**55***a* to **55***g*) are rollers that guide the workpiece A. In this embodiment, each guide roller **55** is a substantially cylindrical roller. In the cooling unit **50**, the conveyance route for the workpiece A is set by the plural guide rollers **55** such that it is directed from the loading section **50***b* (connecting part **72**) toward the outlet (connecting part **74**) through the plural cooling rollers **52**. Among the plural guide rollers **55**, the guide roller **55***b* is a tension detection roller that detects the tension applied to the workpiece A.

[0030] The guide roller **55***a* is disposed such that its uppermost portion is at the same height as the loading section **50***b* of the cooling unit **50**. The guide rollers **55***b* and **55***c* are disposed in a lower portion of the cooling unit **50**. The guide roller **55***b* is disposed below the guide roller **55***a*. The guide roller **55***c* is disposed in front of the guide roller **55***b* (on the right side in the figure). The guide rollers **55***d* and **55***e* are disposed in an upper portion of the cooling unit **50**. The guide roller **55***d* is disposed above the guide roller **55***c*. The guide roller **55***e* is disposed in front of the guide roller **55***d*. The guide rollers **55***f* and **55***g* are disposed near the unloading section **50***c* of the cooling unit **50**. The guide roller **55***g* is disposed such that its lowermost portion is at the same height as the unloading section **50***c* of the cooling unit **50**. The lowermost portion of the guide roller **55***g* is substantially at the same height as the uppermost portion of the guide roller **55***a*. The guide roller **55***f* is positioned higher by one guide roller **55** than the guide roller **55***g*. The guide roller **55***f* is positioned between the guide roller **55***g* and the unloading section **50***c*of the cooling unit **50**.

[0031] The workpiece A is introduced into the cooling unit **50** from the loading section **50***b* of the cooling unit **50** through the guide roller **55***a*. The workpiece A is conveyed substantially horizontally to the cooling unit **50**. The workpiece A is hung on the guide roller **55***a* from above and conveyed downward. The workpiece A is hung on the guide roller **55***b* from the left side and conveyed forward. The workpiece A is hung on the guide roller **55***c* from above and conveyed to the first cooling roller **52***a* disposed at the lowermost end. The workpiece A is then alternately hung around the first cooling rollers **52***a* and the second cooling rollers **52***b* in sequence from the bottom and is conveyed from the second cooling roller **52***b*, which is located at the uppermost end, toward the guide roller **55***d*. The workpiece A is hung on the guide roller **55***d* from below and conveyed forward. The workpiece A is hung on the guide roller **55***e* from above and conveyed downward. The workpiece A is hung on the guide roller **55***f* from the right side and conveyed rearward. The workpiece A is hung on the guide roller **55***g* from above and then folded back forward. The workpiece A is conveyed substantially horizontally toward the unloading section **50***c* of the cooling unit **50**.

[0032] The cooling unit **50** is connected to the winding unit **60** via the connecting part **74**. The connecting part **74** includes the unloading section **50***c* for the cooling unit **50** and an inlet for the winding unit **60**. The cooled workpiece A is conveyed to the winding unit **60** through the connecting part **74**.

Winding Unit **60**

[0033] The winding unit **60** is equipment that winds up the workpiece A. The winding unit **60** houses the winding roll A**2** for winding up the workpiece A that has been cooled through the cooling unit **50**. The winding unit **60** includes the outer wall **61** surrounding internal equipment and the winding roll A**2**. The winding unit **60** is provided with the winding shaft **62** and plural rollers **65**. The winding roll A**2** is attached to the winding shaft **62**, and the workpiece A, which has been heat-treated in the heat treatment unit **40** and cooled in the cooling unit **50**, is wound onto the winding roll A**2**. By rotationally driving the winding shaft **62**, the workpiece A is wound onto the winding roll A**2**.

[0034] Within the space enclosed by the outer wall **61** of the winding unit **60**, the plural rollers **65** that set the conveyance route for the workpiece A are provided. The plural rollers **65** set the conveyance route along which the workpiece A is conveyed within the winding unit **60**. The workpiece A conveyed from the cooling unit **50** is hung over the roller **65** near the inlet (connecting part **74**) of the winding unit **60**, then hung around the plural rollers **65** in the predetermined order, and eventually wound onto the winding roll A**2**. The plural rollers **65** include guide rollers **65***a*, a dancer roller **65***b*, the tension detection roller **65***c*, and feed rollers **65***d*. The dancer roller **65***b* is configured to be movable within a predetermined range. The dancer roller **65***b* can be moved, for example, to secure a necessary extra length of the workpiece A when the winding roll A**2** is replaced. A tension detector (not illustrated) is attached to the tension detection roller **65***c*. The feed roller **65***d* feeds out the necessary extra length of the workpiece A when the workpiece A is attached to a new winding roll A**2** after the replacement of the winding roll A**2**.

Vacuum Pump **80**

[0035] The heat treatment apparatus **10** includes a vacuum pump **80**. Interior spaces of the unwinding unit **30**, the heat treatment unit **40**, the cooling unit **50**, and the winding unit **60** described above are enclosed by the outer walls **31**, **41**, **51**, and **61**, respectively. The unwinding unit **30**, the heat treatment unit **40**, the cooling unit **50**, and the winding unit **60** have the spaces isolated from an external space by the outer walls **31**, **41**, **51**, and **61**, respectively. The interior spaces of the outer walls **31**, **41**, **51**, and **61** communicate with each other when the workpiece A is treated. The vacuum pump **80** is connected to the outer walls **31**, **41**, **51**, and **61** of the respective units. The vacuum pump **80** reduces the pressure in the interior spaces of the unwinding unit **30**, the heat treatment unit **40**, the cooling unit **50** (processing space **50***a* in the cooling unit **50**), and the winding unit **60**. In this embodiment, the workpiece A is processed under a predetermined vacuum atmosphere with a pressure lower than atmospheric pressure.

[0036] The connection form of the vacuum pump **80** is not particularly limited. Plural vacuum pumps **80** may be provided. The respective vacuum pumps **80** may be connected to the unwinding unit **30**, the heat treatment unit **40**, the cooling unit **50**, and the winding unit **60**. Alternatively, pipes may be branched from one vacuum pump **80** to reduce the pressure inside some of the unwinding unit **30**, the heat treatment unit **40**, the cooling unit **50**, and the winding unit **60**.

[0037] The pipe of the vacuum pump **80** is provided with vacuum valves **81** to **84** for adjusting the vacuum level in each unit. The vacuum valves **81** to **84** are configured to be switchable between the connection of each unit to the vacuum pump **80** and the disconnection of each unit from the vacuum pump **80**. In a case where the vacuum level of each unit is not adjusted, open/close valves may be used instead of the vacuum valves **81** to **84**.

[0038] A door **70***a* is provided at the inlet of the heat treatment unit **40** (in this embodiment, the connecting part **70**). The door **70***a* is closed when the unwinding roll A**1** is replaced, for example. By closing the door **70***a* when replacing the unwinding roll A**1** or the like, the atmosphere of the heat treatment unit **40** (in this embodiment, a reduced pressure state) can be maintained. The door **70***a* may be closed when the workpiece A passes through the connecting part **70**, such as when the unwinding roll A**1** is replaced. When the remaining workpiece A wound on the unwinding roll A**1** is nearly depleted, the unwinding roll A**1** is replaced with a new one. An end of a workpiece A on a newly installed unwinding roll A**1** after the replacement is joined to the end of the workpiece A from the previous roll before the replacement. With the workpiece A remaining in the processing space, the unwinding roll A**1** can be replaced while maintaining the atmosphere in the heat treatment unit **40**, allowing for a quick recovery of the apparatus after the replacement of the unwinding roll A**1**.

[0039] A door **74***a* is provided at the unloading section **50***c* of the cooling unit **50** (in this embodiment, the connecting part **74**). Like the door **70***a*, the door **74***a* can maintain the atmosphere (in this embodiment, the reduced pressure state) in the cooling unit **50** by closing the door **74***a* when replacing the winding roll A**2** or the like. The door **74***a* may be closed when the workpiece A passes through the connecting part **74**, such as when the winding roll A**2** is replaced. When the amount of the workpiece A wound on the winding roll A**2** becomes significant, the winding roll A**2** is replaced with a new one. A leading edge of the newly installed winding roll A**2** after the replacement and the end of the workpiece A are joined together. With the workpiece A remaining in the processing space **50***a*, the winding roll A**2** can be replaced while maintaining the atmosphere in the cooling unit **50**, allowing for a quick recovery of the apparatus after the replacement of the winding roll A**2**.

[0040] In the cooling device, a strip-shaped workpiece is cooled while being conveyed along the set conveyance route. The conveyance route is set to be aligned with the plural cooling rollers. The workpiece is loaded from the loading section, hung around the plural cooling rollers, and then unloaded from the unloading section. In the cooling device, the strip-shaped workpiece is cooled by contacting the cooling rollers. When processing the workpiece using such a cooling device, the workpiece is loaded from the loading section, passed between the plural cooling rollers, and discharged through the unloading section in advance, while the cooling device remains stopped. Thereafter, the cooling device is actuated, thereby allowing the workpiece to be cooled while being conveyed. The present inventors intend to improve the workability of causing the workpiece to pass through the plural cooling rollers.

[0041] FIGS. **2** and **3** are schematic diagrams, each illustrating a support structure of the cooling rollers **52**. In FIG. **2**, the support structure of the cooling rollers **52** is illustrated as viewed from the right side. In FIG. **3**, the support structure of the cooling rollers **52** is illustrated as viewed from the left side.

Plural Cooling Roller **52**

[0042] The plural cooling rollers **52** (first cooling rollers **52***a* and second cooling rollers **52***b*) include roller shafts **52***a***1** and **52***b***1** and roller bodies **52***a***2** and **52***b***2**, respectively, as illustrated in FIGS. **2** and **3**. The roller shafts **52***a***1** and **52***b***1** are substantially cylindrical shaft-shaped members. Each of the roller shafts **52***a***1** and **52***b***1** is a member that serves as the rotation axis of the corresponding cooling roller **52**. The roller shafts **52***a***1** and **52***b***1** of the cooling rollers **52** extend in the direction (left-right direction) in which the support plates **53** and **54** in a pair are opposed to each other. Thus, the axial directions of the plural cooling rollers **52** are aligned with one another. The roller shafts **52***a***1** and **52***b***1** are supported by the pair of support plates **53** and **54** via respective bearings **52***c*.

[0043] The roller bodies **52***a***2** and **52***b***2** are attached to the roller shafts **52***a***1** and **52***b***1**, respectively. The roller bodies **52***a***2** and **52***b***2** are members with a substantially cylindrical shape, each having a diameter larger than that of its corresponding roller shaft **52***a***1** or **52***b***1**. The roller bodies **52***a***2** and **52***b***2** are parts that the workpiece A contacts. Within each of the roller bodies **52***a***2** and **52***b***2**, a route is provided to allow the refrigerant to pass therethrough. The route through which the refrigerant passes is joined from the roller shafts **52***a***1** and **52***b***1** to the roller bodies **52***a***2** and **52***b***2**. The roller bodies **52***a***2** and **52***b***2** are cooled by supplying the refrigerant to the roller bodies **52***a***2** and **52***b***2**. The workpiece A is cooled by coming into contact with the cooled roller bodies **52***a***2** and **52***b***2** during conveyance.

[0044] The plural cooling rollers **52** are arranged in a staggered manner, with their positions sequentially displaced along the height direction. Here, of the plural cooling rollers **52**, the first cooling rollers **52***a* are disposed at the first side (front) of the support plates **53** and **54**. Of the plural cooling rollers **52**, the second cooling rollers **52***b* are disposed on the second (rear) side of the support plates **53** and **54**. The first cooling roller **52***a* and the second cooling roller **52***b* are alternately arranged along the height direction.

[0045] The conveyance route for the workpiece A is set such that the workpiece A is loaded from the loading section **50***b* (see FIG. **1**), sequentially hung around the plural cooling rollers **52**, and then unloaded from the unloading section **50***c* (see FIG. **1**). In the conveyance route for the workpiece A, the hanging of the workpiece A is performed so as to bring the first surface A**3** of the workpiece A into contact with the cooling roller (first cooling roller **52***a*) displaced toward the first side and bring the second surface A**4** into contact with the cooling roller (second cooling roller **52***b*) displaced toward the second side, among the plurality of cooling rollers **52**. The workpiece A is alternately hung around the first cooling roller **52***a* and the second cooling roller **52***b*. This increases the length of contact between the workpiece A and the cooling rollers **52**, allowing the workpiece A to be cooled efficiently. The plural cooling rollers **52** are supported by the pair of support plates **53** and **54**.

Pair of Support Parts **53** and **54**

[0046] The support plates **53** and **54** in the pair are opposed to each other. The pair of support plates **53** and **54** is provided in the processing space **50***a* isolated from the outside of the outer wall **51** (see FIG. **1**). The support plates **53** and **54** in the pair are opposed to each other in the axial direction (left-right direction) of the cooling roller **52**. In the processing space **50***a*, the support plate **53** is provided on the right side, while the support plate **54** is provided on the left side. The support plates **53** and **54** in the pair sandwich a space in which the plural cooling rollers **52** are arranged in the staggered manner. In this embodiment, the support plates **53** and **54** in the pair sandwich the space in which the roller bodies **52***a***2** and **52***b***2** are arranged. The pair of support plates **53** and **54** includes the bearings **52***c* that support the respective roller shafts **52***a***1** and **52***b***1** of the plural cooling rollers **52**.

[0047] The pair of support plates **53** and **54** extends along their height direction. The pair of support plates **53** and **54** extends in the height direction up to a position higher than the roller axis **52***b***1** of the cooling roller **52**, which is disposed at the highest point among the plural cooling rollers **52**. The widths (dimensions in the front-back direction) of the support plates **53** and **54** in the pair are set such that the side surfaces of the plural roller bodies **52***a***2** and **52***b***2** are exposed from the pair of support plates **53** and **54** when viewed along the axial direction. In this embodiment, the width of each of the pair of support plates **53** and **54** is narrower than a distance between a position where the roller shaft **52***a***1** of the first cooling roller **52***a* is arranged in the height direction and a position where the roller shaft **52***b***1** of the second cooling roller **52***b* is arranged in the height direction.

[0048] However, the material of the support plates **53** and **54** is not particularly limited. The support plates **53** and **54** can be formed using a stainless steel plate, for example. The thickness and material of the support plates **53** and **54** may be set according to the weight of the plural cooling rollers **52**, the tension applied to the workpiece A during conveyance, and the like.

Support Plate **53**

[0049] The support plate **53** supports right end portions of the cooling rollers **52**. As illustrated in FIG. **2**, through holes **53***c* are formed in the support plate **53**. The shape of the through hole **53***c* is substantially circular. The shape of the through hole **53***c* is not particularly limited and may be polygonal or elliptical. From the viewpoint of the strength of the support plate **53**, the through hole **53***c* preferably has a substantially circular shape. The through hole **53***c* is a working through hole used by workers to perform tasks with their hands or tools during the operation of hanging the workpiece A around the cooling roller **52** (feed work), both before and after the processing of the workpiece A. The dimensions of the through hole **53***c* may be set in consideration of the strength of the support plate **53**. The through hole **53***c* may be set to a dimension that is not too large so that the support plate **53** can withstand the tension applied to the workpiece A during conveyance. Although not particularly limited, the dimension of the through hole **53***c* may be between about 10 cm and about 30 cm, inclusive.

[0050] The through holes **53***c* are formed in the same number as the number of the first cooling rollers **52***a* and also as the number of the second cooling rollers **52***b*. Each of the through holes **53***c* is formed substantially at a center portion of the support plate **53** in the width direction (front-back direction) of the support plate **53**. The through holes **53***c* are formed slightly closer to a side surface **53***a* on the first side. The through holes **53***c* are formed at positions overlapping the conveyance route for the workpiece A in the direction in which the support plates **53** and **54** in the pair (see FIGS. **2** and **3**) are opposed to each other. Here, each through hole **53***c* is provided at a position overlapping the workpiece A, which is hung from the first cooling roller **52***a* to the second cooling roller **52***b*. The workpiece A hung from the first cooling roller **52***a* to the second cooling roller **52***b* overlaps a lower portion of the through hole **53***c*.

[0051] The support plate **53** has support portions **56***a* and **56***b* (first support portions **56***a* and second support portions **56***b*) as illustrated in FIG. **2**. The support portions **56***a* and **56***b* are members that support the cooling rollers **52**. The support portions **56***a* and **56***b* are provided with the bearings **52***c*. The first support portions **56***a* are provided at the side surface **53***a* on the first side. The first support portion **56***a* supports the corresponding first cooling roller **52***a*. The second support portions **56***b* are provided at a side surface **53***b* on the second side. The second support portion **56***b* supports the corresponding second cooling roller **52***b*. The side surface **53***a* on the first side faces forward. The side surface **53***b* on the second side faces rearward. Each of the side surface **53***a* on the first side and the side surface **53***b* on the second side is a flat surface and has a substantially rectangular shape that is longer in the height direction than in the thickness direction (opposed direction).

[0052] The first support portions **56***a* and the second support portions **56***b* protrude from the side surface **53***a* on the first side and the side surface **53***b* on the second side, respectively, of the support plate **53**. Thus, the bearings **52***c* support the cooling rollers **52** in front of and behind the support plate **53**. The first support portion **56***a* protrudes forward relative to the side surface **53***a* on the first side. The second support portion **56***b* protrudes rearward relative to the side surface **53***b* on the second side.

[0053] The first support portions **56***a* and the second support portions **56***b* may be integrally configured with the support plate **53**. The first support portion **56***a* and the second support portion **56***b* may be members that are different from the support plate **53** and may be attached to the support plate **53**. In this embodiment, the first support portion **56***a* and the second support portion **56***b* are separate members from the support plate **53** and are attached to the support plate **53**. The first support portion **56***a* is attached to the side surface **53***a* on the first side of the support plate **53**. The second support portion **56***b* is attached to the side surface **53***b* on the second side of the support plate **53**. Hereinafter, the attachment structure of the first support portions **56***a* to the support plate **53** will be described. Since the attachment structure of the second support portions **56***b* to the support plate **53** is the same as that of the first support portions **56***a* thereto, a detailed description thereof is omitted.

[0054] FIG. **4** is a schematic diagram illustrating the first support portion **56***a*. FIG. **4** illustrates the attachment structure of the support plate **53** and the support portions **56***a* that support the first cooling rollers **52***a*. Each of the first support portions **56***a* is a plate-shaped member with the substantially same thickness as the support plate **53**.

[0055] As illustrated in FIG. **4**, second through holes **53***d* are formed in the support plate **53**. The second through hole **53***d* penetrates the support plate **53** in its thickness direction. The second through hole **53***d* has a smaller diameter than the through hole **53***c*. The second through hole **53***d* is provided in proximity to a position where the first support portion **56***a* is provided. The second through hole **53***d* is provided in proximity to the side surface **53***a* on the first side. Two second through holes **53***d* are provided in upper and lower positions of one first support portion **56***a*, with one second through hole at each position. The upper second through hole **53***d* is provided at a height corresponding to an upper portion of the first support portion **56***a*. The lower second through hole **53***d* is provided at a height corresponding to a lower portion of the first support portion **56***a*.

[0056] The second through hole **53***d* has a substantially rectangular shape with chamfered corners. A third through hole **53***d***2** is formed from a surface **53***d***1** closer to the first support portion **56***a* among the inner circumferential surfaces of the second through hole **53***d*, toward the side surface **53***a* on the first side of the support plate **53**. Attachment holes **56***a***1** are formed in the support portion **56***a*. Two of the attachment holes **56***a***1** are formed in positions corresponding to the upper and lower third through holes **53***d***2**.

[0057] Each first support portion **56***a* is attached to the support plate **53** by attachment members

53*d***3**. For example, a bolt or the like can be used as the attachment member **53***d***3**. The attachment member **53***d***3** is inserted into the third through hole **53***d***2**. The attachment member **53***d***3** penetrates from the inner circumferential surface of the second through hole **53***d* toward the side surface **53***a* on the first side and is attached to the attachment hole **56***a***1**. The attachment hole **56***a***1** can be, for example, a bolt hole into which the attachment member **53***d***3** fits. The first support portion **56***a* has its two portions, i.e., its upper and lower portions attached to the support plate **53**. This allows the first support portion **56***a* to be stably attached to the support plate **53**.
[0058] The first support portion **56***a* and the support plate **53** may be reinforced by a reinforcing plate **53***e*. The reinforcing plate **53***e* is a substantially rectangular stainless steel plate attached across the first support portion **56***a* and the support plate **53**. The material, shape, and the like of the reinforcing plate **53***e* are not particularly limited. The reinforcing plate **53***e* is attached to outer surfaces of the first support portion **56***a* and support plate **53**. The reinforcing plates **53***e* join the respective upper and lower portions of the first support portion **56***a* to the support plate **53**. The provision of the reinforcing plate **53***e* allows the first support portion **56***a* to be stably attached to the support plate **53**. Furthermore, the provision of the reinforcing plate **53***e* also enhances the ability to withstand loads that may be applied to the cooling roller **52** in its axial direction during the processing of the workpiece A or the like.
[0059] In the embodiment described above, the cooling device **50** has the loading section **50***b* (see FIG. **1**), the unloading section **50***c* (see FIG. **1**), the plural cooling rollers **52**, and the pair of support plates **53** and **54**. The loading section **50***b* loads the workpiece A. The unloading section **50***c* unloads the workpiece A. As illustrated in FIGS. **2** and **3**, the support plates **53** and **54** support the plural cooling rollers **52**. The plural cooling rollers **52** are arranged in the staggered manner while being aligned with one another in the axial direction, with their positions sequentially displaced along the height direction. The support plates **53** and **54** in the pair sandwich the space in which the cooling rollers **52** are arranged in the staggered manner. The support plates **53** and **54** each include the bearings **52***c* that support the respective roller shafts **52***a***1** and **52***b***1** of the plural cooling rollers **52**. The conveyance route for the workpiece A is set such that the workpiece A is loaded from the loading section **50***b*, sequentially hung around the plural cooling rollers **52** along the height direction, and then unloaded from the unloading section **50***c*. In addition, the conveyance route for the workpiece A is set so as to bring the first surface A**3** of the workpiece A into contact with the cooling roller **52***a* displaced toward the first side and bring the second surface A**4** of the workpiece A into contact with the cooling roller **52***b* displaced toward the second side, among the plurality of cooling rollers **52**.
[0060] In the cooling device **50**, the conveyance route for the workpiece A is set such that the workpiece A is hung around the plural cooling rollers **52**, which are arranged in the staggered manner along the height direction. The workpiece A is sequentially hung around the plural cooling rollers **52** arranged in the staggered manner along the height direction. It is necessary to sequentially hang the workpiece A on the plural cooling rollers **52** before processing the workpiece A. The cooling rollers **52** are arranged in the space sandwiched between the support plates **53** and **54** in the pair. Thus, when sequentially hanging the workpiece A on the cooling rollers **52**, the workpiece A can be hung on the plural cooling rollers **52** by alternately passing the workpiece A from the outside (front and rear) of the pair of support plates **53** and **54** toward the front side and rear side.
[0061] In the embodiment described above, as illustrated in FIG. **2**, the through holes **53***c* are formed in the support plate **53** of the pair of support plates **53** and **54**. This enables workers to perform an operation of hanging the workpiece A around the cooling rollers **52** with their hands or tools through the through holes **53***c*. As the workpiece A can be easily passed between the cooling rollers **52**, the workability of hanging the workpiece A on the cooling rollers **52** is improved. In addition, the support plate **53** can be made lighter, thus reducing the weight of the entire device.
[0062] The through holes **53***c* are formed at positions overlapping the conveyance route for the

workpiece A in the direction in which the support plates **53** and **54** in the pair are opposed to each other. This makes it easier to access the workpiece A through the through hole **53***c*. Thus, the workability of hanging the workpiece A on the plural cooling rollers **52** can be improved.
[0063] In the embodiment described above, each through hole **53***c* is provided substantially at the center portion of the supply plate **53**. This can improve the workability of passing the workpiece A both when passing the workpiece A from the front cooling roller **52** (in this embodiment, the first cooling roller **52***a*) to the rear cooling roller **52** (in this embodiment, the second cooling roller **52**) and when passing the workpiece A from the rear cooling roller **52** to the front cooling roller **52**.
[0064] However, the through holes **53***c* are formed in one of the pair of support plates **53** and **54** (in this embodiment, the support plate **53** on the right side), but the through hole **53***c* is not limited to such a form. The through hole may also be formed on the other support plate **54**.
[0065] In the embodiment described above, the support plate **53** includes the support portions **56***a* and **56***b* that support the plural cooling rollers **52** at the side surface **53***a* on the first side and the side surface **53***b* on the second side, while the support plate **54** includes the support portions **56***a* and **56***b* that support the plural cooling rollers **52** on a side surface **54***a* on the first side and a side surface **54***b* on the second side. The provision of the support portions **56***a* and **56***b* at the side surfaces **53***a* and **53***b* make it easier for the cooling rollers **52** to be detached from the side surfaces **53***a* and **53***b* of the support plate **53**. Thus, the workability of hanging the workpiece A on the plural cooling rollers **52** can be improved.
[0066] In the embodiment described above, the support portions **56***a* and **56***b* protrude from the side surfaces **53***a* and **54***a* on the first side and the side surfaces **53***b* and **54***b* on the second side, of the pair of support plates **53** and **54**, respectively. Consequently, an area where the cooling rollers **52** overlap the support plate **53** in the axial direction of the cooling roller **52** becomes smaller. As a result, the workability of hanging the workpiece A on the plural cooling rollers **52** can be improved.
[0067] In the embodiment described above, the support plate **53** has the second through holes **53***d* formed in proximity to the positions where the support portions **56***a* and **56***b* are provided. The support portions **56***a* and **56***b* are attached to the support plate **53** by the attachment member **53***d***3** that penetrates from the inner circumferential surface of the second through hole **53***d* toward the closer of the side surface **53***a* on the first side or the side surface **53***b* on the second side. Thus, when attaching and detaching the support portions **56***a* and **56***b* and the cooling rollers **52**, it is easy to work from the outside of the support plate **53** (here, from the right side). As a result, maintainability of the cooling rollers **52**, including replacement, inspection, etc., can be improved.
[0068] In this embodiment, the through holes **53***c* are formed in the support plate **53** of the pair of support plates **53** and **54**. As illustrated in FIG. **3**, the support plate **54** is provided with a refrigerant pipe **91** through which a refrigerant is supplied to the cooling roller **52**.
Support Plate **54**
[0069] The support plate **54** supports left end portions of the cooling rollers **52**. Like the support plate **53**, the support plate **54** includes support portions **56***a* and **56***b* (first support portions **56***a* and second support portions **56***b*) that support the plural cooling rollers **52** at the side surface **54***a* on the first side and at the side surface **54***b* on the second side, respectively. The configuration of each of the support portions **56***a* and **56***b* is the same as that of the support plate **53** described above, and thus a detailed description thereof is omitted.
[0070] Although a detailed illustration is omitted, the refrigerant pipe **91** is joined to each of the ends of the roller shafts **52***a***1** and **52***b***1**. The refrigerant pipes **91** are connected to a refrigerant supply device **90**. The refrigerant supply device **90** supplies the refrigerant toward the refrigerant pipes **91**. The refrigerant is supplied from the roller shafts **52***a***1** and **52***b***1** to the roller bodies **52***a***2** and **52***b***2**, respectively. Each of the cooling rollers **52** is connected to the refrigerant pipe **91**.
[0071] In this embodiment, the cooling roller **52** has a so-called double-tube structure. The double-tube structure is formed at each of the ends of the roller shafts **52***a***1** and **52***b***1**. In the cooling rollers **52**, the refrigerant flows in and out from the left ends of the roller shafts **52***a***1** and **52***b***1**. Each of

the roller shafts **52**_a_**1** and **52**_b_**1** has inner and outer tubes. The refrigerant flows in from the refrigerant pipes **91** connected to the left ends of the roller shafts **52**_a_**1** and **52**_b_**1**, passes between the outer and inner tubes of each of the roller shafts **52**_a_**1** and **52**_b_**1**, passes through the roller bodies **52**_a_**2** and **52**_b_**2**, passes through the inner tube of each of the roller shafts **52**_a_**1** and **52**_b_**1**, and flows out from the refrigerant pipes **91**. The refrigerant pipe **91** may include an inflow pipe that allows the refrigerant to flow into the cooling roller **52** and an outflow pipe that allows the refrigerant to flow out of the cooling roller **52**. However, the configuration of the cooling roller **52** is not limited to the form described above. For example, the flow path for the refrigerant may be set such that the refrigerant circulates among some of the plural cooling rollers **52**.
[0072] Of the pair of support plates **53** and **54**, one support plate **53** is provided with the through holes **53**_c_ (see FIG. **2**). The other support plate **54** is provided with the refrigerant pipes **91** through which the refrigerant is supplied to the cooling rollers **52**. Since the through hole **53**_c_ is provided in the support plate **53**, which is positioned on the side opposite to the refrigerant pipe **91**, the refrigerant pipe **91** does not interfere with the work of hanging the workpiece A on the cooling rollers **52**. This can improve the workability of hanging the workpiece A on the cooling rollers **52**.
[0073] As illustrated in FIGS. **2** and **3**, in the cooling unit **50**, the guide rollers **55**_a_ to **55**_g_ that set the conveyance route for the workpiece A are supported by the above-mentioned pair of support plates **53** and **54**, a pair of support plates **57**_a_ and **57**_b_, and a pair of support plates **58**_a_ and **58**_b_. The pair of support plates **57**_a_ and **57**_b_ is provided on the loading section **50**_b_ side of the cooling unit **50**. The support plates **57**_a_ and **57**_b_ are substantially at the same height as the loading section **50**_b_ and the unloading section **50**_c_. The support plates **58**_a_ and **58**_b_ in a pair are provided to be opposed to each other on the unloading section **50**_c_ side of the cooling unit **50**. The support plates **58**_a_ and **58**_b_ are substantially at the same height as the support plates **53** and **54**, respectively. Each of the support plates **57**_a_ and **58**_a_ is provided substantially on the same plane as the support plate **53**. Each of the support plates **57**_b_ and **58**_b_ is provided substantially on the same plane as the support plate **53**.
[0074] The guide roller **55**_a_ is supported by the pair of support plates **57**_a_ and **57**_b_. The guide roller **55**_b_ is supported by a connecting plate **57**_a_**1** joining the support plate **53** to the support plate **57**_a_, and a connecting plate **57**_b_**1** joining the support plate **54** to the support plate **57**_b_. The guide rollers **55**_c_ and **55**_d_ are supported by the pair of support plates **53** and **54**. The guide rollers **55**_e_ to **55**_g_ are supported by the pair of support plates **58**_a_ and **58**_b_. The right-side support plates **53**, **57**_a_ and **58**_a_ have adjusting members **53**_f_, **57**_c_, and **58**_c_ for adjusting the horizontality of the guide rollers **55**, respectively. The support form of the guide rollers **55** is not limited to such a form.
[0075] The adjacent support plates (here, support plate **53** and support plate **57**_a_, support plate **53** and support plate **58**_a_, support plate **54** and support plate **57**_b_, and support plate **54** and support plate **58**_b_) may be joined together by a detachable frame **59**. The strength of the support plates may be improved by joining the adjacent support plates together through the frame **59**. The frame **59** may be removed, for example, when replacing the cooling roller **52**. The support plates **54** and **57**_a_ may be connected to the detachable frame **59**.
[0076] The above is a detailed description of the technology disclosed herein through the specific embodiments, but those are illustrative only and do not limit the scope of the claims. Accordingly, the technology described in claims includes various variations and modifications of the embodiments described above.
[0077] The present specification includes the following Items 1 to 9. The following Items 1 to 9 are not limited to the above embodiments.
Item 1
[0078] A cooling device including: [0079] a loading section that loads a strip-shaped workpiece; [0080] an unloading section that unloads the workpiece; [0081] a plurality of cooling rollers; and [0082] a pair of support plates that supports the plurality of cooling rollers, wherein [0083] the cooling rollers are arranged in a staggered manner while being aligned with each other in an axial

direction, with positions thereof sequentially displaced along a height direction, [0084] the support plates in the pair are opposed to each other to sandwich a space in which the cooling rollers are arranged in the staggered manner, the support plates including bearings that support respective roller shafts of the plurality of cooling rollers, and [0085] a conveyance route for the workpiece is set such that the workpiece is loaded from the loading section and sequentially hung along the height direction around the cooling rollers so as to bring a first surface of the workpiece into contact with a cooling roller displaced toward a first side and bring a second surface of the workpiece into contact with a cooling roller displaced toward a second side, among the plurality of cooling rollers, and then unloaded from the unloading section.

Item 2

[0086] The cooling device according to Item 1, wherein at least one of the pair of support plates is provided with a through hole.

Item 3

[0087] The cooling device according to Item 2, wherein the through hole is formed at a position overlapping the conveyance route for the workpiece in a direction in which the support plates in the pair are opposed to each other.

Item 4

[0088] The cooling device according to Item 2 or 3, wherein the through hole is provided at a center portion of the support plate.

Item 5

[0089] The cooling device according to any one of Items 2 to 4, wherein [0090] one of the pair of support plates is provided with the through hole, and [0091] the other support plate is provided with a refrigerant pipe through which a refrigerant is supplied to the cooling roller.

Item 6

[0092] The cooling device according to any one of Items 1 to 5, wherein each of the pair of support plates includes support portions that support the plurality of cooling rollers, at a side surface of the support plate on the first side and a side surface of the support plate on the second side.

Item 7

[0093] The cooling device according to Item 6, wherein each of the support portions protrudes from a corresponding one of the side surface on the first side and the side surface on the second side, of each of the pair of support plates.

Item 8

[0094] The cooling device according to Item 6 or 7, wherein [0095] the support plate has a second through hole formed in proximity to a position where the support portion is provided, and [0096] the support portion is attached to the support plate by an attachment member that penetrates from an inner circumferential surface of the second through hole toward a closer one of the side surface on the first side and the side surface on the second side.

Item 9

[0097] A heat treatment apparatus, including: [0098] a heat treatment unit that performs heat treatment on the workpiece; and [0099] the cooling device according to any one of Items 1 to 8 that cools the heat-treated workpiece.

Claims

1. A cooling device, comprising: a loading section that loads a strip-shaped workpiece; an unloading section that unloads the workpiece; a plurality of cooling rollers; and a pair of support plates that supports the plurality of cooling rollers, wherein the cooling rollers are arranged in a staggered manner while being aligned with each other in an axial direction, with positions thereof sequentially displaced along a height direction, the support plates in the pair are opposed to each other to sandwich a space in which the cooling rollers are arranged in the staggered manner, the support plates including bearings that support respective roller shafts of the plurality of cooling rollers, and a conveyance route for the workpiece is set such that the workpiece is loaded from the loading section and sequentially hung along the height direction around the cooling rollers so as to bring a first surface of the workpiece into contact with a cooling roller displaced toward a first side and bring a second surface of the workpiece into contact with a cooling roller displaced toward a second side, among the plurality of cooling rollers, and then unloaded from the unloading section.

2. The cooling device according to claim 1, wherein at least one of the pair of support plates is provided with a through hole.

3. The cooling device according to claim 2, wherein the through hole is formed at a position overlapping the conveyance route for the workpiece in a direction in which the support plates in the pair are opposed to each other.

4. The cooling device according to claim 2, wherein the through hole is provided at a center portion of the support plate.

5. The cooling device according to claim 2, wherein one of the pair of support plates is provided with the through hole, and the other support plate is provided with a refrigerant pipe through which a refrigerant is supplied to the cooling roller.

6. The cooling device according to claim 1, wherein each of the pair of support plates includes support portions that support the plurality of cooling rollers, at a side surface of the support plate on the first side and a side surface of the support plate on the second side.

7. The cooling device according to claim 6, wherein each of the support portions protrudes from a corresponding one of the side surface on the first side and the side surface on the second side, of each of the pair of support plates.

8. The cooling device according to claim 6, wherein the support plate has a second through hole formed in proximity to a position where the support portion is provided, and the support portion is attached to the support plate by an attachment member that penetrates from an inner circumferential surface of the second through hole toward a closer one of the side surface on the first side and the side surface on the second side.

9. A heat treatment apparatus comprising: a heat treatment unit that performs heat treatment on the workpiece; and the cooling device according to claim 1 that cools the heat-treated workpiece.
